

Evaluating Hybrid Retrieval-Augmented Generation for Arabic Regulatory Question Answering: A Comparative Study on Saudi Central Bank Documents

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Abstract

Large language models (LLMs) have demonstrated strong performance in natural language understanding, yet they remain vulnerable to hallucinations and limited access to domain-specific knowledge when answering regulatory questions. This limitation is particularly significant for Arabic financial regulations, where accurate responses require precise interpretation of authoritative legal documents. This study evaluates the effectiveness of a Hybrid Retrieval-Augmented Generation (RAG) architecture against an identical No-RAG baseline for Arabic regulatory question answering using Saudi Central Bank (SAMA) documents.

A hybrid RAG pipeline was developed by combining dense vector retrieval, BM25 lexical retrieval, and Reciprocal Rank Fusion (RRF) with the Aya 8B language model. The system was evaluated against a No-RAG baseline using two manually constructed benchmarks consisting of 50 regulatory questions and a 30-question specificity-controlled dataset. In addition to answer accuracy, retrieval performance was measured using Recall@5, and incorrect RAG responses were classified into retrieval and generation failures.

The RAG system achieved **54.0%** accuracy on the 50-question benchmark compared with **2.0%** for the No-RAG baseline. On the specificity-controlled benchmark, RAG achieved **43.3%** accuracy, outperforming the No-RAG baseline of **13.3%**. Retrieval analysis further showed that relevant regulatory evidence was retrieved for most questions, while generation failures remained a major source of incorrect answers despite successful retrieval.

The findings demonstrate that document-grounded retrieval substantially improves Arabic regulatory question answering under the evaluated experimental conditions, while highlighting the importance of jointly evaluating retrieval quality and generation accuracy in domain-specific RAG systems

1. Introduction

Large Language Models (LLMs) have significantly improved the quality of natural language understanding and generation across a wide range of tasks, including question answering, summarization, and conversational AI. Despite these advances, LLMs remain limited when answering questions that require precise, document-specific knowledge. Their responses are generated primarily from parametric knowledge learned during training, making them susceptible to hallucinations, outdated information, and the inability to reliably reproduce facts contained in authoritative documents. These limitations become particularly important in regulatory

domains, where even minor inaccuracies in numerical values, legal conditions, or reporting requirements can result in incorrect or misleading answers [1].

Retrieval-Augmented Generation (RAG) addresses this limitation by combining a language model with an external knowledge source [1]. Instead of relying solely on internal model knowledge, a RAG system retrieves relevant document passages and supplies them as contextual evidence during generation. Previous research has demonstrated that Retrieval-Augmented Generation (RAG) improves knowledge-intensive question answering by grounding language models in external documents [1]. However, relatively little work has examined the effectiveness of RAG for **Arabic regulatory question answering**, particularly using official financial regulations published by the Saudi Central Bank (SAMA).

Arabic regulatory documents present additional challenges beyond ordinary question answering [2]. The documents contain highly specialized legal terminology, numerical thresholds, licensing conditions, reporting deadlines, and procedural requirements that must be interpreted accurately. Furthermore, regulatory PDFs often require preprocessing before they can be effectively indexed for retrieval, making the quality of document chunking and retrieval an important factor in overall system performance. As a result, evaluating only the final answer accuracy is insufficient; it is equally important to determine whether errors originate from document retrieval or from the language model's interpretation of retrieved evidence.

This paper presents a hybrid RAG framework for Arabic regulatory question answering using a 289-page SAMA financial regulatory corpus [3]. The proposed system combines dense semantic retrieval, BM25 lexical retrieval, and Reciprocal Rank Fusion (RRF) before generating answers with the Aya 8B language model. To evaluate the contribution of retrieval, the system is compared against an identical No-RAG baseline using two manually constructed benchmarks consisting of 50 regulatory questions and a 30-question specificity-controlled dataset. In addition to answer accuracy, the study measures Retrieval Recall@5 and separates RAG errors into retrieval failures and generation failures, providing a more detailed evaluation of document-grounded Arabic question answering.

The main contributions of this work are:

- A hybrid Retrieval-Augmented Generation pipeline designed for Arabic financial regulatory question answering.
- A comparative evaluation between No-RAG and RAG using manually constructed Arabic regulatory benchmarks.
- An error analysis that distinguishes retrieval failures from generation failures rather than evaluating answer accuracy alone.
- An empirical study demonstrating the impact of question specificity on Arabic regulatory QA performance

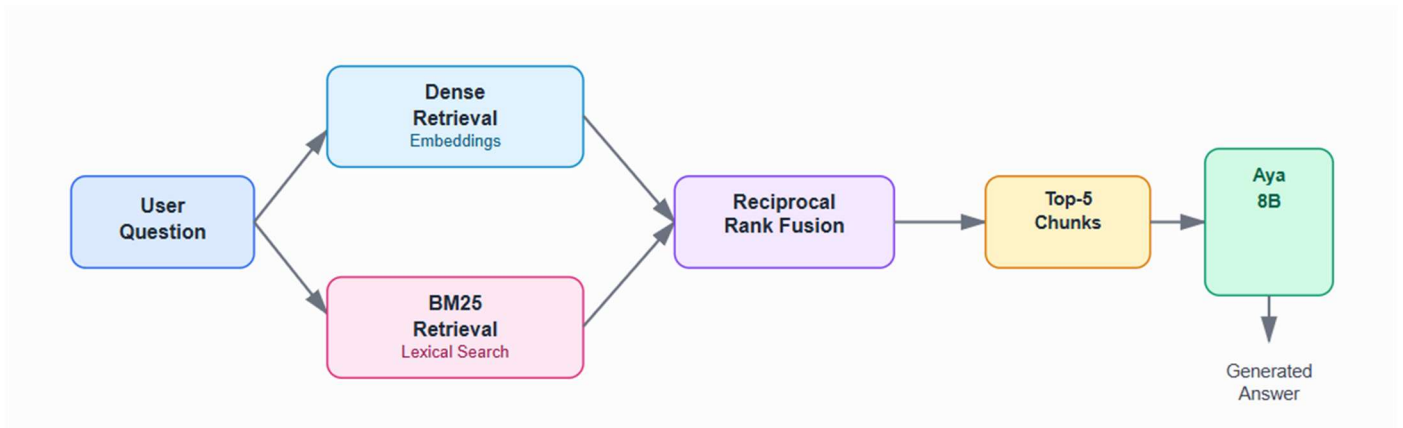
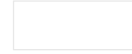


Figure 1. Hybrid RAG Architecture



2. Methodology

This study proposes a Hybrid Retrieval-Augmented Generation (RAG) framework for Arabic regulatory question answering. The system combines dense semantic retrieval, BM25 lexical retrieval, and Reciprocal Rank Fusion (RRF) to retrieve relevant regulatory evidence before generating responses with the Aya 8B multilingual language model. The overall workflow is illustrated in Figure 1.

2.1 Regulatory corpus

The knowledge base consists of a 289-page Arabic financial regulatory document published by the Saudi Central Bank (SAMA). The document contains licensing regulations, compliance requirements, reporting obligations, consumer finance provisions, and other financial-sector policies. The PDF was extracted into plain text and divided into overlapping chunks to preserve contextual continuity during retrieval [3].

2.2 Hybrid retrieval

The retrieval component combines semantic retrieval and lexical retrieval to improve document coverage.

Dense retrieval was implemented using multilingual Sentence-BERT embeddings [4], allowing semantically similar passages to be retrieved even when the wording differs from the user's query.

In parallel, BM25 was used as the lexical retrieval method [5] to capture exact keyword matches and regulatory terminology. BM25 remains one of the most effective probabilistic ranking methods for document retrieval in information retrieval systems.

The rankings produced by both retrieval methods were merged using Reciprocal Rank Fusion (RRF) [6], which assigns higher scores to passages that consistently appear near the top of multiple ranked lists. The five highest-ranked passages were selected as contextual evidence for answer generation.

2.3 Answer generation

Retrieved passages were supplied to the Aya 8B multilingual language model through Ollama [7]. The model was instructed to generate answers exclusively from the retrieved regulatory context, reducing reliance on parametric knowledge and minimizing unsupported responses. Aya 8B was selected because of its strong multilingual capabilities, including Arabic language understanding.

2.4 Experimental design

The proposed Hybrid RAG system was compared against an identical No-RAG baseline. Both systems used the same language model and prompting strategy; the only difference was the availability of retrieved regulatory evidence. Two manually constructed evaluation benchmarks were used: a 50-question regulatory benchmark and a 30-question specificity-controlled benchmark. This controlled design isolates the contribution of retrieval while maintaining identical generation conditions.

3. Results

This section presents the quantitative results of the comparative evaluation between the No-RAG baseline and the proposed Hybrid RAG system. Two independent benchmarks were evaluated: an initial 50-question benchmark and a 30-question specificity-controlled benchmark. The results are reported using answer accuracy, Retrieval Recall@5, and RAG error analysis.

3.1 Overall answer accuracy

The first experiment evaluated both systems on the 50-question Arabic regulatory benchmark. The Hybrid RAG system achieved substantially higher accuracy than the No-RAG baseline, correctly answering 27 of the 50 questions compared with only one correct answer from the baseline.

Benchmark	No-RAG	Hybrid RAG
50 Questions	2.0%	54.0%
30 Questions	13.3%	43.3%

Answer accuracy comparison

Table 1. Answer Accuracy Comparison

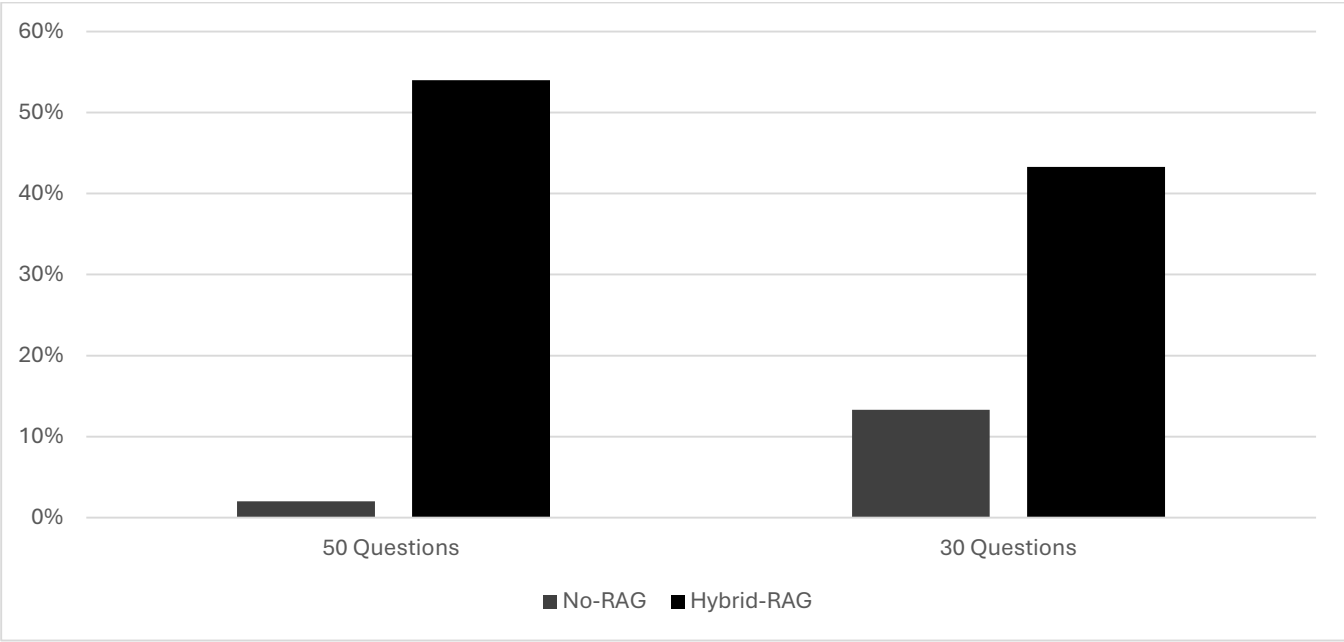


Figure 2. Answer Accuracy Comparison

The second benchmark produced a similar trend. Although the specificity-controlled questions were more challenging, the RAG system maintained a clear advantage over the baseline with an accuracy of 43.3%, while No-RAG achieved 13.3%.

These results demonstrate that incorporating retrieved regulatory evidence consistently improved answer correctness under both evaluation settings.

3.2 Question-level comparison

A question-level analysis was conducted to determine how many questions benefited uniquely from retrieval. For the 50-question benchmark, RAG answered 26 questions correctly that No-RAG failed, while there were no questions in which No-RAG was correct and RAG was incorrect.

Table 2. Question-Level Comparison (50 Questions)

Outcome	Questions
Both systems correct	1
RAG only correct	26
No-RAG only correct	0
Both incorrect	23

For the 30-question specificity-controlled benchmark, RAG produced 11 additional correct answers, whereas No-RAG uniquely answered only two questions correctly.

Table 3. Question-Level Comparison (30 Questions)

Outcome	Questions
Both systems correct	2
RAG only correct	11
No-RAG only correct	2
Both incorrect	15

This comparison indicates that the improvement achieved by RAG was not limited to a small number of questions but was distributed across the benchmark.

3.3 Retrieval performance

Because the RAG system depends on external document retrieval, the retrieval component was evaluated independently using Recall@5. A retrieval was considered successful when the required regulatory evidence appeared within the final five retrieved passages.

The retrieval system successfully provided relevant evidence for the majority of questions in both experiments, although retrieval performance decreased in the specificity-controlled benchmark.

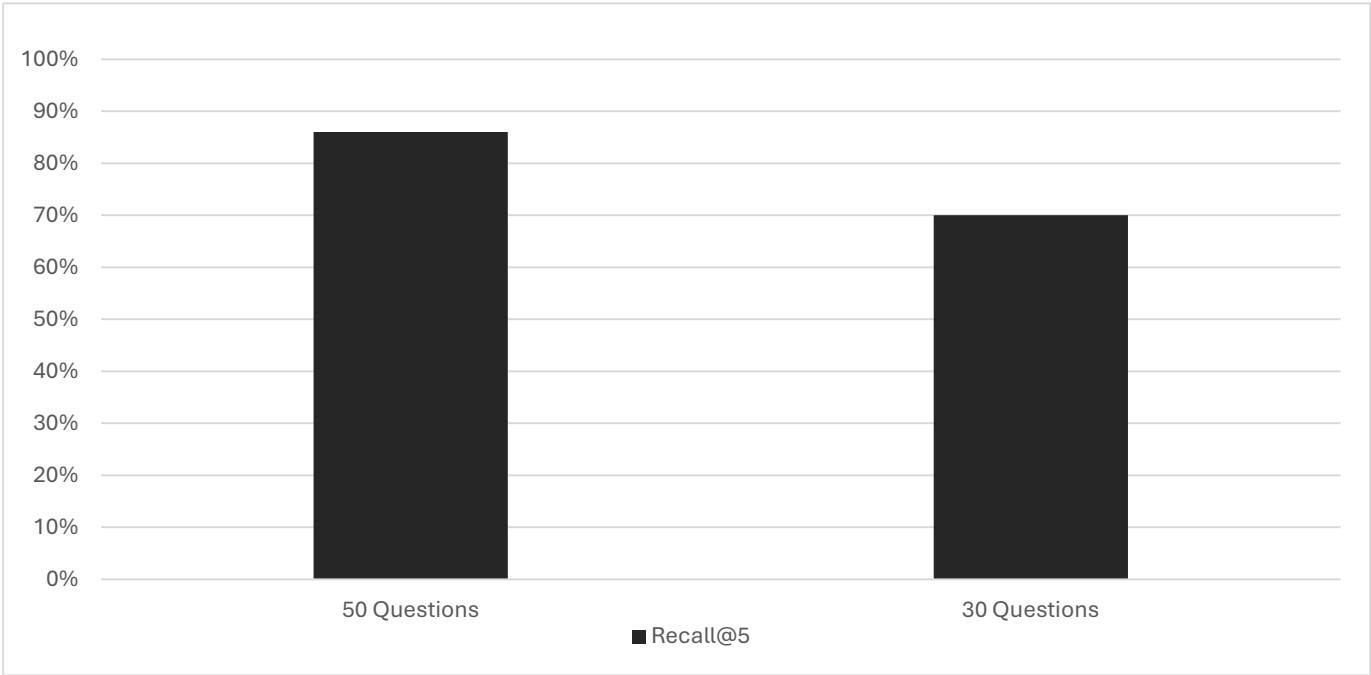


Figure 3. Retrieval Recall@5

3.4 Error analysis

To identify the source of incorrect RAG responses, each error was classified as either a retrieval failure or a generation failure.

A retrieval failure occurred when the required regulatory information was absent from the retrieved context. A generation failure occurred when the correct evidence was present but the language model still produced an incorrect answer.

> 50-question benchmark

RAG error distribution (50-question benchmark)

Breakdown of the 23 incorrect RAG answers.

Table 4. RAG Error Analysis (50-Question Benchmark)

Error Category	Questions	Percentage
Retrieval Failure	6	26.1%
Generation Failure	17	73.9%
Total	23	100%

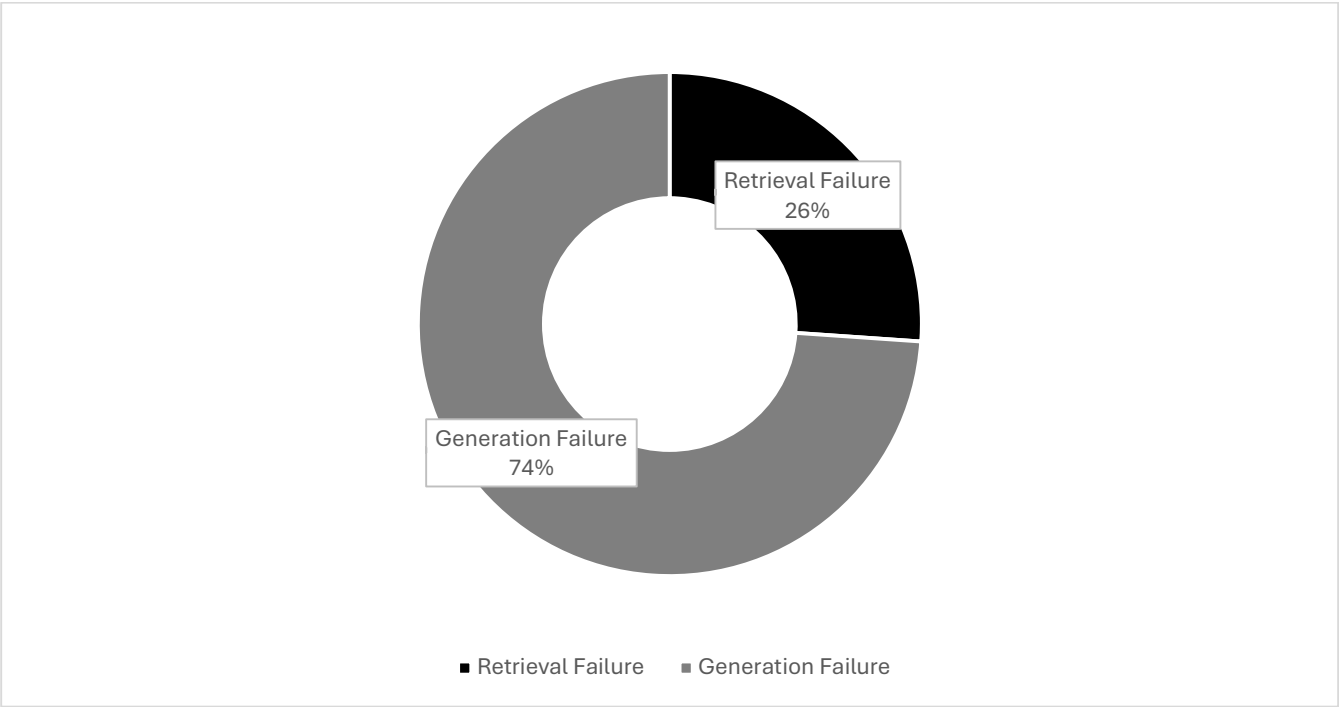


Figure 4. RAG Error Distribution (50Q)

Most incorrect responses occurred after successful retrieval, indicating that the language model's interpretation of retrieved evidence remained the dominant source of error

➔ 30-question specificity-controlled benchmark

RAG error distribution (30-question benchmark)

Breakdown of the 17 incorrect RAG answers.

Table 5. RAG Error Analysis (30-Question Benchmark)

Error Category	Questions	Percentage
Retrieval Failure	8	47.1%
Generation Failure	9	52.9%
Total	17	100%

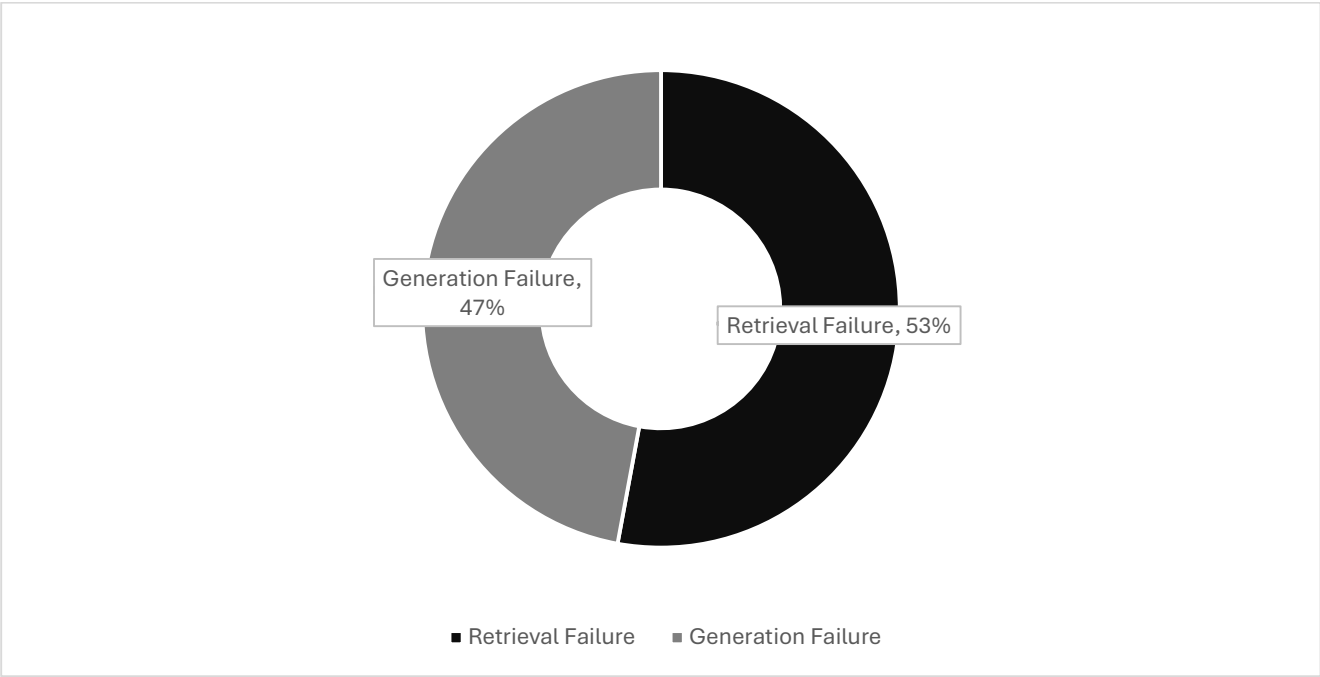


Figure 5. RAG Error Distribution (30Q)

Unlike the original benchmark, retrieval and generation errors were nearly balanced, suggesting that retrieval became a more significant limiting factor in the specificity-controlled evaluation.

3.5 Summary of experimental results

The overall findings from both experiments are summarized below.

Table 6 — Overall Experimental Summary

Metric	50 Questions	30 Questions
No-RAG Accuracy	2.0%	13.3%
Hybrid RAG Accuracy	54.0%	43.3%
Accuracy Improvement	+52.0 pp	+30.0 pp
Retrieval Recall@5	86.0%	70.0%
Generation Failures	17	9
Retrieval Failures	6	8

4. Discussion

The experiments demonstrate that Retrieval-Augmented Generation (RAG) substantially improves Arabic regulatory question answering under the evaluated experimental conditions. While the No-RAG baseline relied entirely on the language model's internal knowledge, the Hybrid RAG system was able to ground its responses using authoritative SAMA regulatory documents, resulting in consistently higher answer accuracy across both evaluation benchmarks. However, the results also reveal that retrieval alone is insufficient for reliable regulatory QA, as a considerable proportion of errors occurred even when relevant evidence had been successfully retrieved.

4.1 Impact of Retrieval on Regulatory Question Answering

The largest performance improvement was observed in the original 50-question benchmark, where the Hybrid RAG system increased accuracy from 2.0% to 54.0%. This suggests that many regulatory questions depend on document-specific information that is unlikely to be reproduced accurately from the language model's parametric knowledge alone.

Financial regulations frequently contain exact numerical values, eligibility conditions, licensing periods, reporting deadlines, and procedural requirements. Unlike general factual questions, these answers are typically defined by a specific regulatory provision rather than common world knowledge. Providing retrieved passages therefore allows the language model to generate responses that are grounded in the source document instead of relying on memorized or inferred information.

The improvement observed in the specificity-controlled benchmark further supports this conclusion, although the magnitude of improvement was smaller. Explicitly identifying the regulatory activity reduced ambiguity in question interpretation and produced more consistent comparisons between the two systems.

4.2 Retrieval Success Does Not Guarantee Correct Answers

One of the most important findings of this study is the distinction between retrieval quality and generation quality.

The retrieval component achieved high Recall@5, indicating that relevant regulatory evidence was available for the majority of questions. Nevertheless, overall answer accuracy remained substantially lower than retrieval performance. This gap demonstrates that successfully retrieving the correct evidence does not necessarily ensure that the language model will correctly interpret or reproduce that evidence.

The error analysis showed that many incorrect RAG responses were classified as generation failures rather than retrieval failures. In these cases, the required regulatory provision was present in the retrieved context, yet the model selected an incorrect numerical value, omitted an important condition, or generated an unsupported statement. This finding highlights an important limitation of current RAG systems: document grounding reduces hallucination, but it does not eliminate reasoning or interpretation errors.

4.3 Influence of Question Specificity

The second benchmark was introduced to examine whether question specificity influences regulatory QA performance. Unlike the original benchmark, every question explicitly identified the relevant regulatory activity or domain.

Although RAG continued to outperform the No-RAG baseline, retrieval performance decreased from 86.0% to 70.0% Recall@5. This suggests that clearer question wording does not automatically produce easier retrieval. Instead, retrieval effectiveness also depends on how closely the terminology used in the question matches the language and structure of the regulatory document.

This observation indicates that benchmark construction is an important factor when evaluating domain-specific QA systems. Question ambiguity and document terminology should therefore be considered alongside model performance when designing Arabic regulatory evaluation datasets.

4.4 Practical Implications

The findings suggest that Hybrid RAG provides a practical approach for Arabic regulatory question answering, particularly in domains where answers must be supported by authoritative documents. However, the remaining retrieval and generation errors indicate that such systems should not be used as fully autonomous regulatory advisors without additional verification mechanisms.

Future systems could improve reliability by incorporating citation-aware generation, stronger reranking models, larger multilingual embedding models, or post-generation verification against the retrieved evidence. These approaches may reduce the generation failures observed in this study while preserving the benefits of document-grounded retrieval.

Overall, this research demonstrates that evaluating RAG should extend beyond final answer accuracy. Separating retrieval failures from generation failures provides a clearer understanding of where the system succeeds and where further improvements are required, making it a useful evaluation framework for future Arabic regulatory QA research.

5. Conclusion

This study presented a Hybrid Retrieval-Augmented Generation (RAG) framework for Arabic regulatory question answering using financial regulations published by the Saudi Central Bank (SAMA). The proposed system combined dense semantic retrieval, BM25 lexical retrieval, and Reciprocal Rank Fusion (RRF), and was evaluated against an identical No-RAG baseline using two manually constructed Arabic regulatory benchmarks.

The experimental results demonstrated that incorporating retrieved regulatory evidence significantly improved answer accuracy under the evaluated conditions. The Hybrid RAG system achieved 54.0% accuracy on the 50-question benchmark compared with 2.0% for the No-RAG baseline, while the specificity-controlled benchmark showed an improvement from 13.3% to 43.3%. These findings indicate that document-grounded retrieval substantially enhances the ability of a language model to answer domain-specific regulatory questions.

Beyond overall accuracy, this research showed the importance of evaluating retrieval quality and generation quality separately. The error analysis revealed that incorrect responses originated from both retrieval failures, where relevant evidence was not retrieved, and generation failures, where the model misinterpreted evidence that was successfully retrieved. This distinction provides a more informative evaluation framework than answer accuracy alone and helps identify where future improvements should be directed.

The study contributes an empirical benchmark for Arabic regulatory QA, a reproducible hybrid RAG pipeline, and a structured methodology for separating retrieval and generation errors. These contributions provide a foundation for future research on Arabic document-grounded question answering in highly specialized domains.

Future work may extend this research by evaluating larger multilingual language models, exploring alternative embedding and reranking techniques, expanding the regulatory corpus to multiple SAMA documents, and incorporating citation-aware or evidence-verification mechanisms to improve the reliability of regulatory question answering systems.

References

- [1] P. Lewis, E. Perez, A. Piktus et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” in *Advances in Neural Information Processing Systems (NeurIPS 2020)*, Vancouver, Canada, 2020.
- [2] M. Jarrar, N. Habash and F. Al-Shargi, “Arabic Natural Language Processing: An Overview,” *Journal of King Saud University – Computer and Information Sciences*, vol. 33, no. 5, 2021.
- [3] Saudi Central Bank (SAMA), “Finance Sector,” Riyadh, Saudi Arabia, 2024.
- [4] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings Using Siamese BERT-Networks,” in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP-IJCNLP)*, Hong Kong, China, 2019.
- [5] S. Robertson and H. Zaragoza, “The Probabilistic Relevance Framework: BM25 and Beyond,” in *Foundations and Trends in Information Retrieval*, Now Publishers, 2009.
- [6] G. V. Cormack, C. L. A. Clarke and S. Buettcher, “Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods,” in *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, Boston, Massachusetts, USA, 2009.
- [7] V. Aryabumi, J. Dang, D. Talupuru et al., “Aya 23: Open Weight Releases to Further Multilingual Progress,” 2024.